

"Dave, this conversation can serve no purpose any more. Goodbye." Thus ran the chilling conversation between the AI program HAL 9000 and the space explorer Dave, moments before Dave breathes his last. Are software programs smart enough to be compared with HAL from Stanley Kubrick's *Space Odyssey 2001*?



Making up for the lack of a brain in Dr Frankenstein's monster, scientists are busy tinkering with creations that can think. A computer program that closely mirrors the manner in which humans learn in the social menagerie of life. These new portents of intelligence use a technique called inductive logic programming—knowledge supplied by the scientists in the form of rules and rule-execution criteria. The programs use this information to learn from their system in what can be termed as 'experience'; and for their own benefit, tweak and change the base rules. It is an evolutionary process that relies on feedback from the environment.

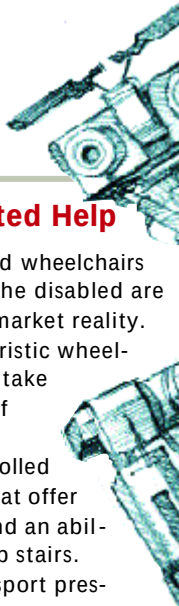
So what are the benefits of such programs? Researchers have already assigned them the role of molecular

biologists: the analysis of protein structure. In one such scenario, the programs successfully 'self-discovered' a previously unknown carcinogen. They were also able to extrapolate and extract new rules that govern the manner in which a three-dimensional protein structure is formed. Success has not been restricted to the field of biology—mathematics was gifted a new sequence of integers called refractorables, by a similarly thinking software routine.

A thinking piece of software is well in demand in various fields, ranging from military—a smart bomb that designates and destroys enemy targets; to medicine—a smart pill that finds home in your body and delivers drugs based upon the feedback it receives from your immune and circulatory system.

Automated Help

Mechanised wheelchairs that help the disabled are already a market reality. These futuristic wheelchairs can take the form of motorised, gyro-controlled vehicles that offer stability and an ability to climb stairs. They also sport pressure-sensitive manoeuvring systems that can be used to steer the machines with the faintest of blows from the mouth. Experts at Osaka University in Japan have used AI techniques to take the current-generation of wheelchairs onto a higher level of automation. You can call the chair towards you, for example by simply gesturing with your hands. Video cameras mounted on the machine along with image recognition and motion detection software allow just such behaviour. The chair can be steered with sideways head motions.



The 1986 Disney film, *Flight of the Navigator*, showcased the story of a young boy taken on an interstellar ride by a sleek silver spacecraft that called itself MAX. When can we expect our current modes of flight to go sentient and fly us around the universe?

Using a combination of data collected from radars, satellites and wireless networks, an airplane is more aware of itself and its airspace than it has ever been. These systems act as early-warning alarms, automated co-pilots, navigational aids and even landing assists. For instance, when a rudder or the wing flaps of an airplane malfunction, there is precious little that a pilot can do to avert disaster, since these systems do not have fallbacks. AI routines developed by NASA have been used to keep a plane flying even when it otherwise wouldn't. The software is based on a neural network—an AI routine that enables a program to perform a series of tasks by learning through trial and error. A flight control system keeps an eye on data such as the plane's airspeed, heading and the forces experienced by its elements such as the wings and the tail. This data is then compared to a safe set of numbers, which would be encountered when things are working optimally. When

something awry is detected by the control system, the neural net kicks in and takes control. The network sifts through the alarming data six times a second to work out the best means of using the operational systems of the plane to maintain normal flight. For example, if it detects a wing-flap, the net would guide the airplane into a pre-programmed manoeuvre, maybe banking the entire structure to compensate for the flaps. The neural net was able to keep the plane in control even at a supersonic speed.

A neural net, along with surface recognition algorithms and data gathered from satellites can theoretically, fly and control an aeroplane. Although such a system might never find widespread use, scenarios wherein the pilots are not available would be invaluable in disasters, like they currently do. Hostage situations, in particular, could be foiled, as a neural net would guide and land the plane in a safe territory without needing human input.